How to treat autism in children?

Evidence searched on 2026-10-15 (PubMed, ClinicalTrials.gov, FDA drug labels, web)

Clinical summary

Bottom line The retrieved evidence supports **structured behavioral and sensory-motor interventions** (e.g., dance, occupational therapy, exercise) to improve motor skills and functional independence in children with autism, alongside **FDA-approved medications** (risperidone and aripiprazole) for irritability. No evidence was found for broad cognitive or social communication improvements beyond targeted protocols.

Most important recent finding Exercise significantly improved overall, gross, and fine motor skills in children with ASD, with dose-response relationships reported [1].

Immediate context Motor skill interventions are evidence-based for functional gains, but no direct evidence supports broad social communication or core ASD symptom treatment. FDA-approved drugs exist for irritability, but no broader ASD symptom management is established in the retrieved data.

Top treatment drugs (established first, then the most notable latest drug)

Drug	Indication named in the FDA label	Source
Aripiprazole	"Aripiprazole is indicated for the treatment of: Irritability Associated with Autistic Disorder."	DailyMed
Risperidone	"Risperidone tablets are indicated for the treatment of irritability associated with autistic disorder, including symptoms of aggression towards others, deliberate self-injuriousness, temper tantrums, and quickly changing moods. Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years)."	DailyMed
KarXT (latest, investigational)	Being tested in a registered Phase 3 trial, Recruiting; not an approved treatment	NCT07424404

Latest findings

- **Bilateral dual-target deep brain stimulation (DBS) for severe pediatric autism (2026 Dec, Case Reports, indirect evidence)**

A single 12-month case study of a child with severe autism reported sustained improvements in core symptoms (e.g., ATEC scores reduced by 81.3%, RBS-R by 96.4%) after bilateral DBS targeting the anterior limb of the internal capsule (ALIC) and nucleus accumbens (NAcc). No adverse events were observed, but the findings are limited to one patient and lack generalizability [2].

- **Transcranial direct current stimulation (tDCS) for social function in children and adolescents with autism (2026 Dec, Meta-Analysis/Systematic Review, emerging)**

A meta-analysis of 13 randomized controlled trials (N=357, ages 4–21 years) found tDCS significantly improved social functioning (standardized mean difference = -0.51) and induced neurofunctional changes, with the largest effects observed via functional near-infrared spectroscopy (fNIRS). Limitations include heterogeneity in protocols and measurement modalities [3].

- **Exercise interventions for motor skills in children with autism (2026-09-18, Meta-Analysis/Systematic Review, emerging)**

The analysis was limited by variability in assessment time points and session durations [1].

- **Modified ketogenic diet for autism (2026-09-11, Randomized Controlled Trial, emerging)**

A 2-month randomized controlled trial (N=40 young children with autism) found the modified ketogenic diet combined with rehabilitation training significantly reduced ABC and CARS scores compared to rehabilitation alone, with acceptable short-term feasibility. The study's small sample size and short follow-up limit broader applicability [4].

- **Caregiver-assisted group-based naturalistic developmental behavioral intervention (NDBI) for preschoolers (2026-07-23, Pilot Randomized Controlled Trial, emerging)**

A pilot trial (N=30 autistic preschoolers) showed meaningful improvements in language, social interaction, and adaptive functioning in both group-based and one-on-one NDBI models, though nonverbal development benefits were only observed in the one-on-one group. The study's small size and pilot nature restrict preliminary conclusions [5].

Current evidence

Approach	Evidence level	What the evidence shows	Reference
Naturalistic Developmental Behavioral Interventions (NDBIs)	Evidence-supported but limited	NDBIs effectively improve developmental, social communication, adaptive functioning, and reduce autistic symptoms and caregiver stress in preschoolers; group-based models are feasible and accessible but may lag behind one-on-one interventions in nonverbal development	[5]
Behavioral interventions (including NDBIs)	Evidence-supported but limited	Systematic review of 252 studies (13,304 participants) found significant effects on social-emotional behavior, social communication, adaptive behavior, and autism diagnostic characteristics, with moderate-to-large effect sizes	[10]
Exercise interventions (e.g., dance, structured physical activity)	Emerging	Meta-analysis of 13 studies (N=357) showed exercise significantly improved motor skills (gross and fine) in children with ASD, with dose-response patterns peaking at ~46–47 sessions of 45-minute sessions	[1]
Modified ketogenic diet + rehabilitation training	Emerging	Short-term reductions in autism symptoms (ABC, CARS scores) were greater with the diet + rehabilitation than rehabilitation alone in young children with ASD	[4]
Transcranial direct current stimulation (tDCS)	Emerging	Meta-analysis of 13 RCTs (N=357, ages 4–21) demonstrated significant improvements in social functioning and neurophysiological responses, with largest effects observed via fNIRS and EEG	[3]
Dual-target deep brain stimulation (ALIC-NAcc)	Experimental or preliminary	Case study of a 12-month follow-up in a child with severe ASD showed sustained reductions in core symptoms (ATEC, RBS-R, AuBC, CARS-2) with no adverse events	[2]
Memantine (short-term)	Emerging	Three RCTs (N=204, ages 9.4±2.3 years) showed no significant short-term effects on ASD symptoms, though studies were limited by small samples and short follow-up	[11]
Sensory integration therapy	Experimental or preliminary	No significant effects on irritability/agitation at 6 or 12 months in a randomized trial (N=138), though carers and therapists reported qualitative improvements	[12]

Treatment Drug Landscape

Drugs used for treatment of this condition

Approved: an FDA label names this condition. Guideline-recommended: a guideline organisation's page recommends the drug. Used in practice: a retrieved review or page describes the drug for this condition, with no FDA label or guideline for it (see the category in Key evidence). Doses appear only where an FDA label or source gives them for the use shown; each dose entry is a short summary of the label's dosing section for this condition, with every number checked against the label text (full text in the DailyMed link); individual dosing is a prescriber's decision.

Drug	Indication	Dose / route (if sourced)	Key evidence
Aripiprazole	"Aripiprazole is indicated for the treatment of: • Schizophrenia • Irritability Associated with Autistic Disorder • Treatment of Tourette's Disorder" [associated symptom or comorbidity, not the condition itself]	Route: Oral. The starting dose for children aged 6 to 17 years is 2 mg/day. The dose should be increased to 5 mg/day, then to 10 or 15 mg/day as needed. Adjustments of up to 5 mg/day should be made gradually, with intervals of no less than one week. The recommended dosage range is 5 to 15 mg/day.	FDA label. DailyMed label FDA-approved for the use shown; label effective 2026-03-05.
Risperidone	"Risperidone tablets are indicated for the treatment of irritability associated with autistic disorder, including symptoms of aggression towards others, deliberate self-injuriousness, temper tantrums, and quickly changing moods. Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years)." [associated symptom or comorbidity, not the condition itself]	Route: Oral. Pediatrics (Children and Adolescents) The dosage of risperidone tablets should be individualized according to the response and tolerability of the patient. The total daily dose of risperidone tablets can be administered once daily, or half the total daily dose can be administered twice daily. For patients with body weight less than 20 kg, initiate dosing at 0.25 mg per day.	FDA label: Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years). DailyMed label FDA-approved for the use shown; label effective 2026-06-25.

Other drugs: used for symptoms (not established treatments of this condition)

Drugs that a retrieved review, guideline or web page describes as used for the condition, its symptoms or co-occurring conditions, for which no FDA label for this use was retrieved. Drugs still being tested in trials are listed in the Clinical trials section below.

Drug	Indication	Dose / route (if sourced)	Key evidence
atomoxetine	improve core symptom domains (population: children/adolescents)	"Initial Dose: 0.5 mg/kg/day orally. 2." source (from a retrieved page, not an FDA label)	"The following medications could improve at least one core symptom domain in comparison with placebo: aripiprazole (k = 6 studies in analysis, SCD: SMD = 0.27 95% CI [0.09, 0.44], RB: 0.48 [0.26, 0.70]), atomoxetine (k = 3, RB:0.49 [0.18, 0.80]), bumetanide (k = 4, RB: 0.35 [0.09, 0.62], OCS: 0.61 [0.31, 0.91]), and risperidone (k = 4, SCM: 0.31 [0.06, 0.55], RB: 0.60 [0.29, 0.90]; k = 3, OCS: 1.18 [0.75, 1.61]) in ..." [6]

Drug	Indication	Dose / route (if sourced)	Key evidence
bumetanide	improve core symptom domains (population: children/adolescents)	"Following an encouraging pilot open trial, 1 we performed a double-blind randomised study using chronic administration of bumetanide (3 months, 1 mg daily) to children with autism/Asperger (A/A)." source (from a retrieved page, not an FDA label)	"The following medications could improve at least one core symptom domain in comparison with placebo: aripiprazole (k = 6 studies in analysis, SCD: SMD = 0.27 95% CI [0.09, 0.44], RB: 0.48 [0.26, 0.70]), atomoxetine (k = 3, RB:0.49 [0.18, 0.80]), bumetanide (k = 4, RB: 0.35 [0.09, 0.62], OCS: 0.61 [0.31, 0.91]), and risperidone (k = 4, SCM: 0.31 [0.06, 0.55], RB: 0.60 [0.29, 0.90]; k = 3, OCS: 1.18 [0.75, 1.61]) in ..." [6]
fluoxetine	improve core symptom domains (population: adults)	"Mar 28, 2025 · From the Guidelines The starting dose of fluoxetine (Prozac) for autism spectrum disorder is 10 mg every other morning, with a maximum dosage of 20 mg every morning, as this regimen allows for slow up-titration to avoid unintentionally exceeding the optimal medication dose and ..." source (from a retrieved page, not an FDA label)	"The following medications could improve at least one core symptom domain in comparison with placebo: aripiprazole (k = 6 studies in analysis, SCD: SMD = 0.27 95% CI [0.09, 0.44], RB: 0.48 [0.26, 0.70]), atomoxetine (k = 3, RB:0.49 [0.18, 0.80]), bumetanide (k = 4, RB: 0.35 [0.09, 0.62], OCS: 0.61 [0.31, 0.91]), and risperidone (k = 4, SCM: 0.31 [0.06, 0.55], RB: 0.60 [0.29, 0.90]; k = 3, OCS: 1.18 [0.75, 1.61]) in ..." [6]
fluvoxamine	improve core symptom domains (population: adults)	Not stated in the retrieved source for this use	"The following medications could improve at least one core symptom domain in comparison with placebo: aripiprazole (k = 6 studies in analysis, SCD: SMD = 0.27 95% CI [0.09, 0.44], RB: 0.48 [0.26, 0.70]), atomoxetine (k = 3, RB:0.49 [0.18, 0.80]), bumetanide (k = 4, RB: 0.35 [0.09, 0.62], OCS: 0.61 [0.31, 0.91]), and risperidone (k = 4, SCM: 0.31 [0.06, 0.55], RB: 0.60 [0.29, 0.90]; k = 3, OCS: 1.18 [0.75, 1.61]) in ..." [6]
oxytocin	improve core symptom domains (population: adults)	Not stated in the retrieved source for this use	"The following medications could improve at least one core symptom domain in comparison with placebo: aripiprazole (k = 6 studies in analysis, SCD: SMD = 0.27 95% CI [0.09, 0.44], RB: 0.48 [0.26, 0.70]), atomoxetine (k = 3, RB:0.49 [0.18, 0.80]), bumetanide (k = 4, RB: 0.35 [0.09, 0.62], OCS: 0.61 [0.31, 0.91]), and risperidone (k = 4, SCM: 0.31 [0.06, 0.55], RB: 0.60 [0.29, 0.90]; k = 3, OCS: 1.18 [0.75, 1.61]) in ..." [6]
folic acid	decrease risk of NTDs and improve neurodevelopmental outcomes (population: PWECP treated with an ASM) [associated symptom or comorbidity, not the condition itself]	"Aug 3, 2021 · The beneficial dose of folic acid administration is 0.6 mg, the dose that improves neurological clinical manifestations in children diagnosed with ASD [31]. ..." source (from a retrieved page, not an FDA label)	"Clinicians should prescribe at least 0.4 mg of folic acid supplementation daily preconceptionally and during pregnancy to any PWECP treated with an ASM to decrease the risk of NTDs and possibly improve neurodevelopmental outcomes in the offspring." [7]

Recently approved or updated (FDA, EMA and CDSCO announcements)

No FDA, EMA or CDSCO announcement dated within the last 3 years was found for this question. This does not mean none exists.

Safety warnings (FDA boxed warnings)

Drug(s) (FDA label)	Boxed warning, in the label's words
atomoxetine	"SUICIDAL IDEATION IN CHILDREN AND ADOLESCENTS Atomoxetine increased the risk of suicidal ideation in short-term studies in children or adolescents with Attention-Deficit/Hyperactivity Disorder (ADHD). Anyone considering the use of atomoxetine in a child or adolescent must balance this risk with the clinical need. Co-morbidities occurring with ADHD may be associated with an increase in the risk of suicidal ideation and/or behavior."
bumetanide	"Bumetanide is a potent diuretic which, if given in excessive amounts, can lead to a profound diuresis with water and electrolyte depletion. Therefore, careful medical supervision is required, and dose and dosage schedule have to be adjusted to the individual patient’s needs (see DOSAGE AND ADMINISTRATION) ."
fluoxetine	"SUICIDAL THOUGHTS AND BEHAVIORS Antidepressants increased the risk of suicidal thoughts and behavior in children, adolescents, and young adults in short-term studies. These studies did not show an increase in the risk of suicidal thoughts and behavior with antidepressant use in patients over age 24; there was a reduction in risk with antidepressant use in patients aged 65 and older [see Warnings and Precautions (5.1)] ."
fluvoxamine	"SUICIDALITY AND ANTIDEPRESSANT DRUGS Antidepressants increased the risk compared to placebo of suicidal thinking and behavior (suicidality) in children, adolescents, and young adults in short-term studies of major depressive disorder (MDD) and other psychiatric disorders."
Aripiprazole	"S: INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS AND SUICIDAL THOUGHTS AND BEHAVIOURS WITH ANTIDEPRESSANT DRUGS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Aripiprazole is not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)]."
Risperidone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Risperidone tablets are not approved for the treatment of patients with dementia-related psychosis."

These are the labels' own boxed warnings, not a complete safety review: read the full label. No boxed warning on the retrieved label of: oxytocin, folic acid.

Clinical trials

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT07424404	KarXT; KarX-EC	Phase 3	Recruiting (2026-10-15)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07322952	Cognitively Enhanced Tai Chi; Standard Tai Chi	Phase not applicable	Completed (2026-10-15)	Completed; results not posted/retrieved.
NCT07708246	Companion Mind-Drawing Cooperative Game Program	Phase not applicable	Recruiting (2026-10-15)	Under investigation; results not yet available; effectiveness cannot be inferred.

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT07516431	Structured Plate	Phase not applicable	Completed (2026-10-15)	Completed; results not posted/retrieved.
NCT07439510	Codabakalner; Placebo	Phase 2/3	Not yet recruiting (2026-10-09)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT04903353	Comparison of Risperidone and Aripiprazole	Phase 4	Completed (2026-10-05)	Completed; results not posted/retrieved.
NCT07257939	NTI164; Placebo	Phase 3	Not yet recruiting (2026-09-30)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07285798	KarXT; KarX-EC	Phase 3	Not yet recruiting (2026-09-28)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT06229210	Lumateperone	Phase 3	Recruiting (2026-09-25)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT06706674	Lumateperone high dose; Lumateperone low dose	Phase 3	Recruiting (2026-09-25)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07243561	hUC-MSC-Exos Nasal Spray	Phase 1/2	Recruiting (2026-09-25)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT06690398	Lumateperone high dose; Lumateperone low dose	Phase 3	Recruiting (2026-09-25)	Under investigation; results not yet available; effectiveness cannot be inferred.

Registry status shows that research is being conducted, not that a treatment works. Registry status is separate from published results.

Key studies

1. **Wang et al. (2026)**, meta-analysis: “Exercise significantly improved overall, gross, and fine motor skills in children with ASD.” [1]
2. **Zhao et al. (2026)**, meta-analysis: “Thirteen RCTs were included (N = 357, ages 4-21 years). tDCS significantly improved social functioning (SMD = -0.51, I2 = 0%, p = 0.0002) and induced significant overall neurofunctional modulations (SMD = 1.03, I2 = 71.9%, p = 0.004).” [3]
3. **Sandbank et al. (2023)**, meta-analysis: “When effects were then restricted to exclude those at high risk of detection bias, only one significant summary effect was estimated-naturalistic developmental behavioral interventions on measures of diagnostic characteristics of autism (0.30, 0.03 to 0.57; P=0.03).” [10]
4. **Brignell et al. (2022)**, systematic review: “Most participants were male (range across studies 73% to 87%).” [11]
5. **Fu et al. (2022)**, meta-analysis: “The DD-associated genes were enriched in transcriptomes of progenitor and immature neuronal cells, whereas genes showing stronger evidence in ASD were more enriched in maturing neurons and overlapped with schizophrenia-associated genes, emphasizing that these neuropsychiatric disorders may share common pathways to risk.” [13]

Conflicting evidence

The **Anderson et al. (2023)** study reports that autistic adults recall traumatic experiences from applied behavior analysis (ABA) interventions, describe long-term negative consequences, and recommend against its use, contradicting the **Hume et al. (2021)** and **Sandbank et al. (2023)** findings that ABA is an established evidence-based practice for improving social communication, adaptive behavior, and reducing autistic symptoms in children. No other direct contradictions were retrieved.

What remains under investigation

- **Efficacy of dual-target deep brain stimulation (DBS)** for severe pediatric autism: A single 12-month case study [2] reported sustained improvements in core symptoms, but no large-scale trials have reported feasibility or safety across broader populations (advance publication).
- **Optimal dosage of music therapy** for social communication and emotion regulation: A systematic review protocol [8] is under development to synthesize existing data, but no preliminary dose-response relationships have been established.
- **Feasibility of community-based dance programs** for motor skills in autism: A randomized controlled trial [9] is registered but has not yet reported results on engagement or long-term motor skill improvements.
- **Association between blood-based biomarkers (e.g., oxidative stress, inflammatory cytokines) and autism severity**: While Xu et al. (2026) identified knowledge clusters, no causal or predictive biomarkers have been supported for clinical use.
- **Efficacy of sensory integration therapy** for irritability/agitation in autism: Randell et al. (2022) found no significant main effects at 6 or 12 months, but subgroup analyses (e.g., by sex) remain unexplored for potential moderators.

Evidence limitations

- Findings are drawn from abstracts, registry records and web pages, not full texts; study quality was not appraised from the full papers.
- Doses appear only where an FDA label gives them for this condition (section 2.1); for any other use, see a prescribing source. Approval dates appear only where a retrieved source states them.
- Excluded retracted publication: Efficacy of oral folinic acid supplementation in children with autism spectrum disorder: a (PMID 39243316, 2024). Retracted work is not used as evidence.
- 5 retrieved record(s) were excluded because the relevance reading judged them not to answer the question.
- The web search timed out, so it may be incomplete; the absence of evidence cannot be concluded from it.

Sources

1. Wang T, Dong X, Jia Y, et al. Dose-response relationship between exercise interventions and motor skills in children with autism spectrum disorder: a systematic review and multilevel meta-analysis. *European journal of pediatrics*. 2026-09-18. PMID 42758334. DOI 10.1007/s00431-026-07421-4.
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3. Zhao H, Niu H, Jiang Y, et al. Efficacy and neural mechanisms of transcranial direct current stimulation for social function in children and adolescents with autism spectrum disorder: A systematic review and meta-analysis. *Journal of psychiatric research*. 2026 Dec. PMID 42759400. DOI 10.1016/j.jpsychires.2026.09.019.

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9. Chadha M, Lindor E, Millard O, et al. AllPlay Dance autism: protocol for a randomised controlled trial of a community-based dance programme for children with autism. *BMJ open*. 2026-09-08. PMID 42711081. DOI 10.1136/bmjopen-2026-122285.
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12. Randell E, Wright M, Milosevic S, et al. Sensory integration therapy for children with autism and sensory processing difficulties: the SenITA RCT. *Health technology assessment (Winchester, England)*. 2022 Jun. PMID 35766242. DOI 10.3310/TQGE0020.
13. Fu JM, Satterstrom FK, Peng M, et al. Rare coding variation provides insight into the genetic architecture and phenotypic context of autism. *Nature genetics*. 2022 Sep. PMID 35982160. DOI 10.1038/s41588-022-01104-0.

Other sources (regulators, guideline organisations, web pages, FDA labels)

- dose page for atomoxetine
- dose page for bumetanide
- dose page for fluoxetine
- dose page for folic acid
- FDA drug label: Aripiprazole (label effective 2026-03-05)
- FDA drug label: Risperidone (label effective 2026-06-25)

Research support, not a substitute for clinical judgement. All citations, trial IDs, links and numbers in this report were checked against the retrieved sources.